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How To Set \$PATH Permanently in Linux

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An essential part of Linux and other Unix-like operating systems is the \$PATH variable. It instructs the shell where to look for these executable files and specifies a list of directories that house various executables on the system. The folders in \$PATH on a brand-new Linux installation ensure that most of the programs you run function flawlessly. On your system, you might occasionally want to quickly run customized scripts or apps. In these circumstances, you must adequately specify the \$PATH variable before executing those apps.

What Is \$PATH in Linux?

As was already established, the Linux shell will refer to the folders listed in the \$PATH environment variable when it receives a command or an exec call from a program. The \$PATH variable typically contains the directories /bin, /usr/bin, and /usr/local/bin, with superuser receiving two additional entries, namely /sbin and /usr/sbin. To simply start scripts and programs from the terminal regardless of the current working directory, you can add additional entries to your \$PATH variable.

To see the current PATH setting:

```
echo $PATH
```

How to Set the \$PATH Variable in Linux?

There are various ways to add a directory to your \$PATH, depending on what you want to do, whether it's a temporary configuration to run a script only once or you plan to use the script frequently.



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- Method 1 – Setting \$PATH Variable Temporarily
- Method 2 – Setting a Permanent \$PATH Variable

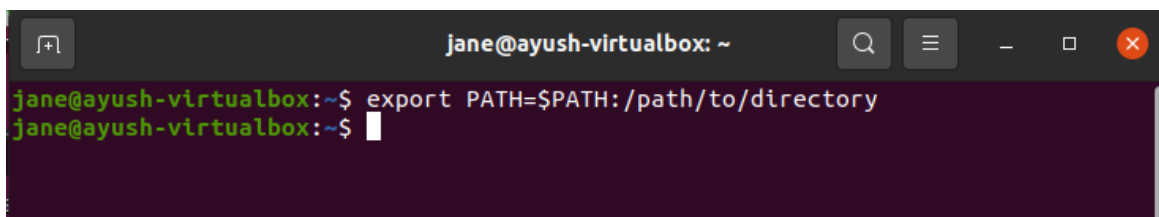
Method 1: Setting \$PATH Variable Temporarily

Setting a temporary \$PATH variable is an option if you just want to run a script or program within the current (active) session. You will then be able to run that program with a command from anywhere on your system and only in the current session, rather than needing to include its complete path in the command.

Syntax:

Open the terminal and enter the following syntax to add folders to the temporary \$PATH variable:

```
export PATH=$PATH:/path/to/directory
```

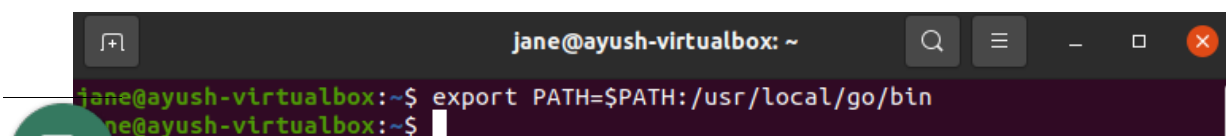
A terminal window titled 'jane@ayush-virtualbox: ~' with a search icon, menu icon, and window control buttons. The prompt is 'jane@ayush-virtualbox:~\$'. The command 'export PATH=\$PATH:/path/to/directory' has been entered and executed, with a new prompt 'jane@ayush-virtualbox:~\$' appearing on the next line.

```
jane@ayush-virtualbox:~$ export PATH=$PATH:/path/to/directory
jane@ayush-virtualbox:~$
```

Example:

To set the temporary path of the **go** program we use the below cmd.

```
export PATH=$PATH:/usr/local/go/bin
```

A terminal window titled 'jane@ayush-virtualbox: ~' with a search icon, menu icon, and window control buttons. The prompt is 'jane@ayush-virtualbox:~\$'. The command 'export PATH=\$PATH:/usr/local/go/bin' has been entered and executed, with a new prompt 'jane@ayush-virtualbox:~\$' appearing on the next line.

```
jane@ayush-virtualbox:~$ export PATH=$PATH:/usr/local/go/bin
jane@ayush-virtualbox:~$
```

Note: Keep in mind that with this setup, you can only run the program during the current session. After you restart your computer, the system will revert the \$PATH variable you temporarily set to its default value.

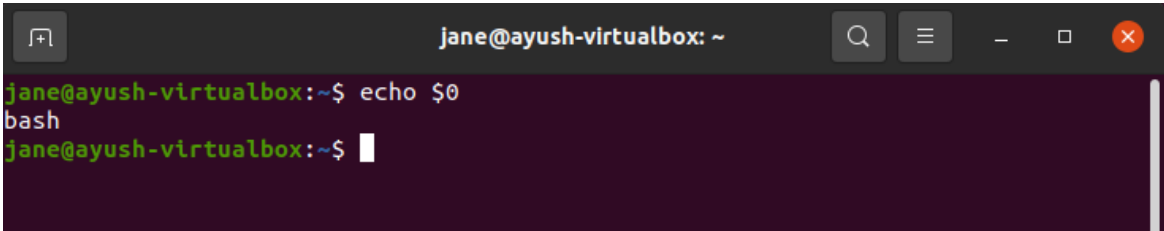
Method 2: Setting a Permanent \$PATH Variable

We must permanently configure the \$PATH variable for any software on your system that you anticipate using regularly. This will ensure that your shell will remember its directory even after a reboot. Setting \$PATH permanently, however, entails a few extra steps in contrast to the temporary solution:

Find the configuration file for the shell you’re using, change it, and add the \$PATH variable there.

Step 1: Run the following command in the terminal to determine your system’s shell:

```
echo $0
```



The output should show the name of your current shell. Unless you’ve changed it to Zsh or another Linux shell, this will often be the Bash shell.

We must now edit the configuration file for the shell you are using. Depending on your shell, you should change the following file:

Shell Name	Configuration File
1. bash	~/.bashrc

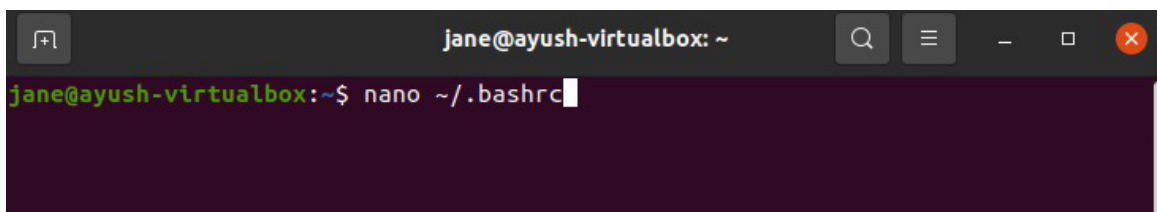


Shell Name	Configuration File
3. ksh	~/.kshrc
4. csh	~/.cshrc

The methods below will help you set the \$PATH variable after you've identified the file that needs editing. In this tutorial, I'll show you how to configure the \$PATH variable in the Bash shell.

Step 2: Open the terminal and enter the following command to open the .bashrc file.

```
nano ~/.bashrc
```



```
jane@ayush-virtualbox: ~$ nano ~/.bashrc
```

Syntax:

Use the following syntax to set the \$PATH variable.

```
export PATH=$PATH:/path/to/directory/
```

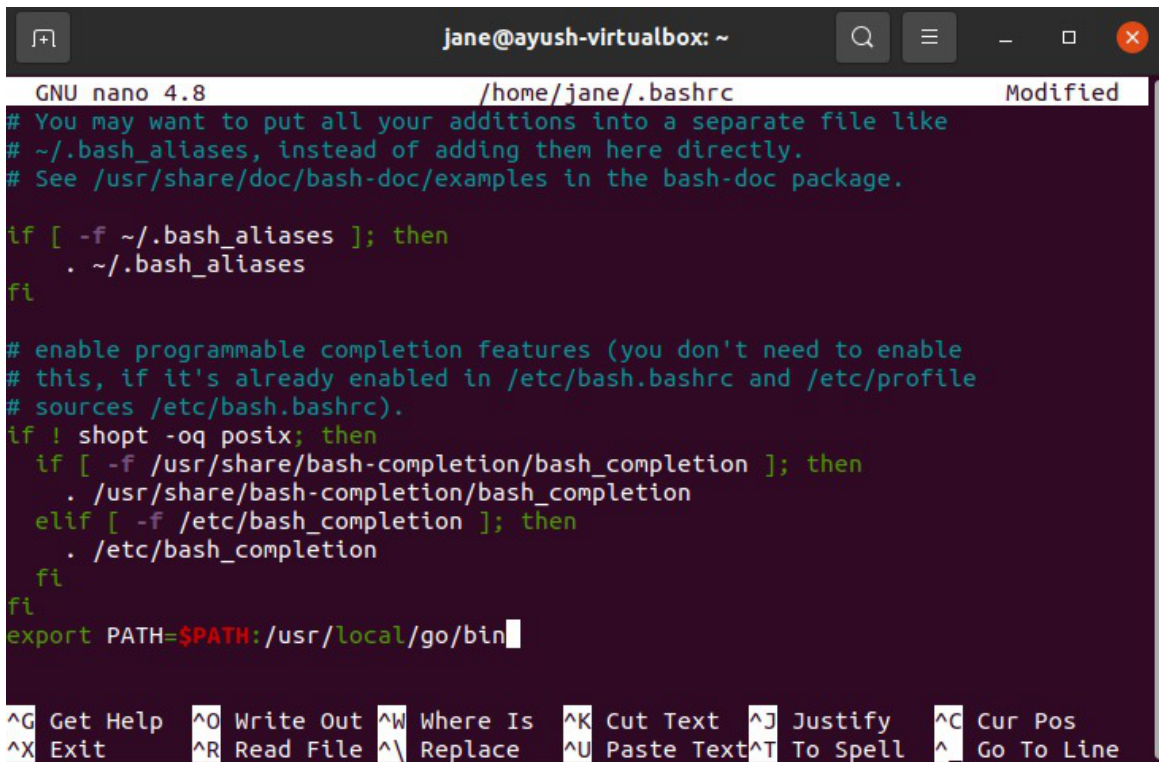
Example:

Here We have added the path of the go program at the end of the .bashrc file.



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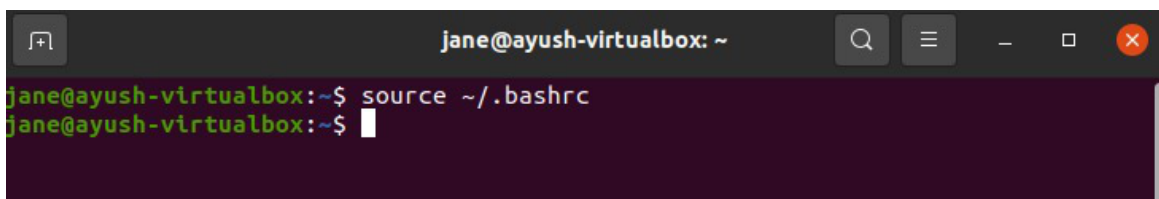


```
GNU nano 4.8 /home/jane/.bashrc Modified
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

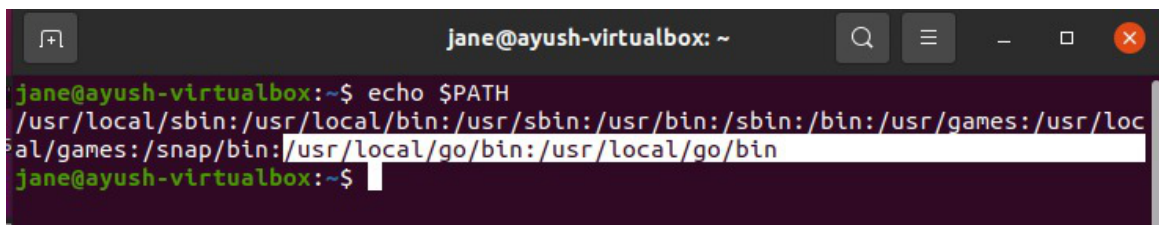
# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
    if [ -f /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -f /etc/bash_completion ]; then
        . /etc/bash_completion
    fi
fi
export PATH=$PATH:/usr/local/go/bin
```

Step 3: Save the file and reload the current shell environment using the source.



```
jane@ayush-virtualbox:~$ source ~/.bashrc
jane@ayush-virtualbox:~$
```

Step 4: Type the following command to check whether the directory has been added or not.



```
jane@ayush-virtualbox:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/usr/local/go/bin:/usr/local/go/bin
jane@ayush-virtualbox:~$
```

All the scripts in the folder should be executable from any directory on your system if you input the directory path correctly, which will appear in your system's \$PATH. Furthermore, Linux gives you the option to permanently configure the \$PATH variable in a system-wide configuration

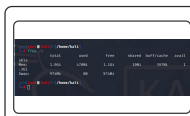
scripts that are located in the given directory. The same command you used to add temporary and permanent setup must be added to the `/etc/environment` or `/etc/profile` file if you want to set `$PATH` globally.

Open either of these files and append the path of the directory:

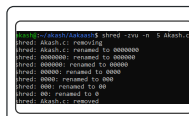
```
nano /etc/environment
nano /etc/profile
```

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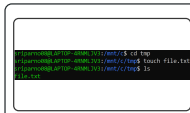
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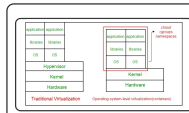
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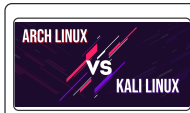
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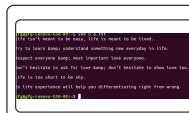
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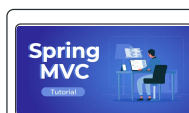


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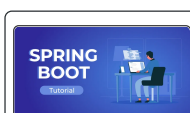
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